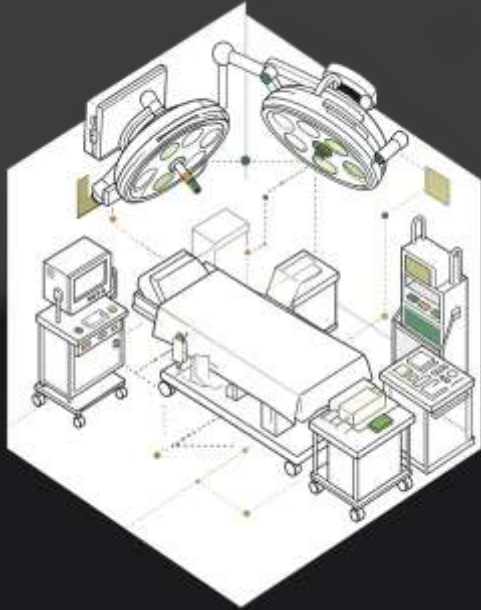




Hospital Bed & Operating Room Scheduling

Subject: DSA 3

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INTRODUCTION



Hospitals have limited resources such as beds, operating rooms, medical staff, and time slots.



Managing these resources manually can lead to delays, conflicts, and inefficient resource usage.



Patients may have different priorities, treatment durations, and deadlines.



This project develops a scheduling system that assigns patients to suitable beds, operating rooms, staff, and available time slots.



The system aims to improve resource utilization, reduce patient waiting time, and minimize scheduling conflicts.

Abstract

Hospital Bed and Operating Room Scheduling is a challenging task because hospitals have limited beds, operating rooms, staff, and time slots, while patients may have different priorities and treatment requirements. This project focuses on developing a scheduling engine that helps hospitals assign patients and medical procedures to suitable beds, operating rooms, and available time slots. The system considers factors such as procedure duration, patient priority, staff availability, room capacity, and treatment deadlines while creating the schedule. It aims to reduce unnecessary waiting time, room idle time, and scheduling conflicts, while making better use of available hospital resources. The system also provides quality reports showing important measures such as room utilization, patient waiting time, and overall scheduling efficiency. By automating the scheduling process, the project can help hospital staff manage resources more effectively and make faster scheduling decisions. Overall, the system provides a practical approach to improving hospital operations and patient flow.



Software & Hardware Requirements

Software Requirements

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Programming Language

Java.

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Core Tools

CodeTantra, HackerRank.

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Integrated Development Environment(IDE)

Visual Studio Code (VS Code).

Hardware Requirements

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Processor

Intel Core i3 / AMD
Ryzen 3 or higher

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Memory

Minimum 4 GB RAM; 8
GB recommended

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Storage

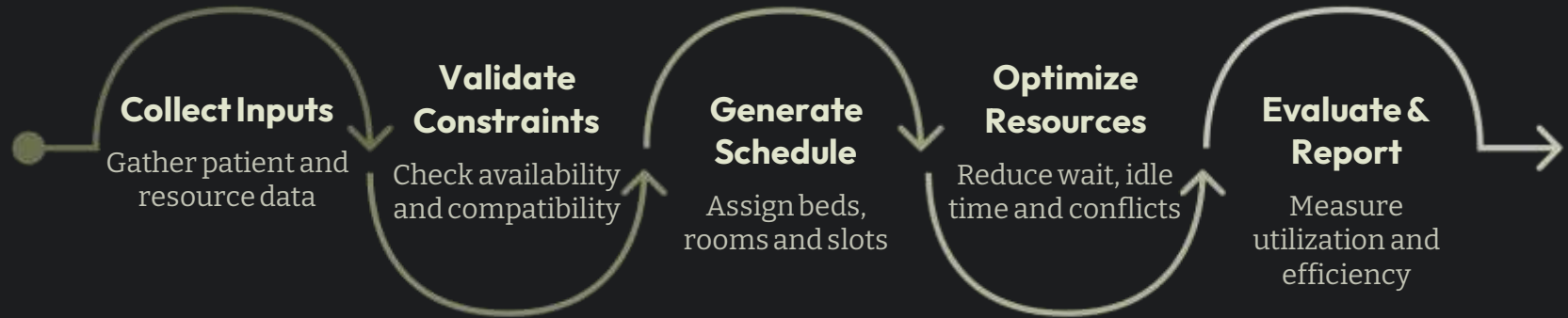
At least 10 GB available
disk space

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Peripherals

Standard keyboard,
mouse, monitor, and
network connection

Methodology



The scheduling engine processes raw patient and resource data through a structured pipeline — from input collection and constraint validation to schedule generation, optimization, and final reporting — enabling hospital staff to make fast, informed decisions.

Literature



Study/Area	Main Focus	Limitation
Hospital Scheduling Systems	Automated patient and resource scheduling	May not handle multiple constraints
Operating Room Scheduling	Optimizing operating room usage	Often focuses mainly on ORs
Priority-Based Scheduling	Gives preference to high-priority patients	Can increase waiting time for lower-priority patients
Optimization Techniques	Reduces idle time and improves utilization	Can be computationally complex
Proposed System	Combines beds, ORs, staff, priorities, and time slots	Designed as a practical DSA-based solution

Thank You

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